Comparative Analysis of Regression Tree Models for Premises Valuation Using Statistica Data Miner

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Abstract. Several dozen of experiments were conducted with Statistica Data Miner in order to assess the suitability of different machine learning algorithms for an Internet expert system to assist with real estate appraisal. The investigations concentrated first of all on regression trees and ensemble tree models. Moreover, decision tree approaches were compared with commonly used algorithms as KNN, SVM and a multilayer perceptron neural network. The results provided by the collection of twelve predictive accuracy measures were also analyzed. The study proved the usefulness of majority of algorithms to build the real estate valuation models.

Keywords: regression trees, machine learning, real estate appraisal, Statistica.

1 Introduction

The application of trees in discrimination and regression analysis was introduced by Breiman et al. in 1984 [1]. Shortcut of the title of their book (CART) gave the name to the whole family of methods and algorithms. New methods and algorithms referring to trees i.e. CHAID, M5, boosted trees, MAR Splines, forests and others were developed in the course of time. Integrating regression trees with other regression approaches is also alternative use of trees which overcome the limitations of individual methods both in terms of accuracy as well in computational efficiency [2]. Trees found their application in many areas of life, e.g. in economy [3], industry [4], mechanics [5], computer science [6], mathematics [7], medicine [8], robotics [9], real estate appraisal [10]. The results presented in tree-form are intuitive, aid decision making and are easy to use. Interesting subsets of models can be indentified along with their characteristics by following the path from the subset to the root of the tree. The technique is fairly robust to the presence of outliers, is stable with highly uncorrelated data, and can handle missing values [11]. Regression trees can be learned from large datasets, and they perform better on larger data sets than on smaller ones [12], [13].

In our pervious works [14], [15] we investigated different machine learning algorithms, among others genetic fuzzy systems and artificial neural networks, devoted to

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build data driven models to assist with real estate appraisals using MATLAB and KEEL. In this paper, which is the continuation of our research, we report the results of experiments conducted with Statistica Data Miner aimed at the comparison of several decision tree methods using actual data taken from cadastral system in order to assess their appropriateness to an Internet expert system assisting appraisers' work.

2 Cadastral Systems as the Source Base for Model Generation

The concept of a data driven models for premises valuation, presented in the paper, was developed basing on sales comparison method. It was assumed that whole appraisal area, that means the area of a city or a district, is split into sections (e.g. clusters) of comparable property attributes. The architecture of the proposed system is shown in Fig. 1. The appraiser accesses the system through the internet and chooses an appropriate section and input the values of the attributes of the premises being evaluated into the system, which calculates the output using a given model. The final result as a suggested value of the property is sent back to the appraiser.

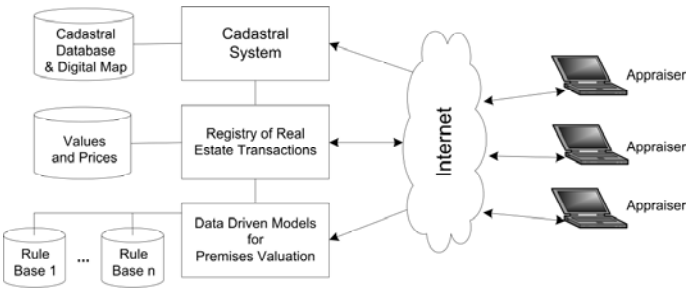

Fig. 1. Information systems to assist with real estate appraisals

Actual data used to generate and learn appraisal models came from the cadastral system and the registry of real estate transactions referring to residential premises sold in one of big Polish cities at market prices within two years 2001 and 2002. They constituted original data set of 1098 cases of sales/purchase transactions. Four attributes were pointed out as price drivers: usable area of premises, floor on which premises were located, year of building construction, number of storeys in the building, in turn, price of premises was the output variable.

3 Regression Algorithms Used in Experiments

In our study we used nine machine learning algorithms implemented in Statistica, one of the most popular statistic and analytics software developed by StatSoft (www.statsoft.com). Statistica Data Miner can be characterized by some useful features including a comprehensive selection of data mining methods, e.g., clustering techniques, neural networks architectures, classification/regression trees, multivariate modeling, and many other predictive techniques; a large selection of graphics and

visualization procedures. Moreover, it provides complete data mining projects, ready to run and set up to competitively evaluate alternative models, using bagging, boosting, stacking, meta-learning, etc., and to produce presentation-quality summary reports [16]. The algorithms are listed in Table 1, they comprise four algorithms to construct single regression trees (CRT, CHI, ICR, IEC), two to build ensemble tree models (BST, FOR), and three other popular machine learning algorithms (KNN, ANN, and SVM).

Table 1. Statistica machine learning algorithms used in study

Code	Description
CRT	C&RT algorithm - Classification and Regression Trees
CHI	CHAID tree - Chi-squared Automatic Interaction Detector
ICR	Interactive C&RT algorithm
IEC	Interactive Exhaustive CHAID algorithm
BST	Boosting trees
FOR	Random forests
KNN	K-Nearest Neighbors with Euclidean distance measure
ANN	Artificial Neural Networks with MLP architecture
SVM	Support Vector Machines with RBF core

CRT. C&RT algorithm is based on binary recursive partitioning. At each node, which represents a group - subsets of the training data, there is applied binary rule which produces two branches. Such process, which divides input spaces into regions, is repeated till the given threshold is reached or when the algorithm finds out that no further gain is made.

CHI. CHAID algorithm builds non-binary trees, i.e., trees where more than two branches can attach to a single root or node. Construct trees, where each non-terminal node identifies a split condition, to yield optimum prediction of continuous dependent variables. It is well suited for the analysis of larger datasets.

ICR. Applying this algorithm user can add and remove nodes and whole levels of trees. User have full control in process of building tree. User can in few seconds make a tree and judge its quality.

IEC. This modification to the basic CHAID algorithm performs a more thorough merging and testing of predictor variables, specifically, the merging of categories continues until only two categories remain for each predictor. The algorithm then proceeds the split variable step, and selects among the predictors the one that yields the most significant split.

BST. The algorithm evolved from the application of boosting methods to regression trees. The general idea is to compute a sequence of simple trees, where each successive tree is built for the prediction residuals of the preceding tree. This method builds binary trees, i.e., partition the data into two samples at each split node.

FOR. It is an implementation of the Random Forest algorithm developed by Breiman [17]. A Random Forest consists of an ensemble of simple tree classifiers, each capable of producing a response when presented with a set of predictor values.

KNN. It is a memory-based method that requires no training. Predictions are based on a set of prototype examples that are used to predict new (i.e., unseen) data based

on averaging over a set of k-nearest prototypes. In this algorithm Euclidean measure was employed for measuring the distance between the query point and cases from the examples sample.

ANN. During network training, the weights and thresholds are first initialized to small, random values. This implies that the units' response surfaces are each aligned randomly with low slope: they are effectively uncommitted. As training progresses, the units' response surfaces are rotated and shifted into appropriate positions, and the magnitudes of the weights grow as they commit to modeling particular parts of the target response surface.

SVM. This method performs regression tasks by constructing nonlinear decision boundaries. It assumes that the relationship between the independent and dependent variables is given by a deterministic function f plus the addition of some additive noise. The task is then to find a functional form for f that can correctly predict new cases that the SVM has not been presented with before. In this algorithm as kernel the radial basis function (RBF) was used.

4 Plan of Experiments

The goal of the research was to carry out comparative analysis of several regression tree algorithms implemented in the Statistica Data Miner and a few the most popular machine learning regression algorithms. Some preliminary tests were accomplished in order to determine the best parameters of respective algorithms using trial and error method.

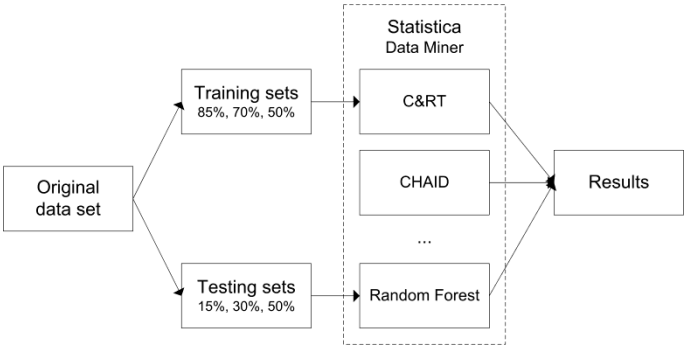

Fig. 2. Schema of the experiments with Statistica Data Miner

Schema of the experiments comprising algorithms with preselected parameters is depicted in Fig. 2. Three different random splits of the original data set into training and testing sets were performed. The latter contained 15%, 30% and 50% of cases included in the original data set, that means 152, 337, and 582 cases respectively. In order to avoid the impact of an unfavourable drawing, the random splits were repeated 5 times for each case and as averaging function median was applied. Normalization of data was accomplished using the min-max approach. As fitness function the mean absolute percentage error (MAPE) was applied and cross validation available in

Statistica was employed. SVM, ANN and KNN were run using 25-fold cross validation, in turn, trees except BST and FOR were run using 15-fold cross validation. A dozen of commonly used performance measures [18], [19] was applied to evaluate models built by respective algorithms used in our study. These measures are listed in Table 2 and expressed in the form of following formulas below, where y_i denotes actual price and $\hat{y}_i$ – predicted price of i-th case, *avg(v), var(v), std(v)* – average, variance, and standard deviation of variables $v_1, v_2, \ldots, v_N$, respectively and N – number of cases in the testing set.

Table 2. Performance measures used in study

Denot.	Description	Dimension	Min value	Max value	Desirable outcome	No. of form.
MSE	Mean squared error	d^2	0	∞	min	1
RMSE	Root mean squared error	d	0	∞	min	2
RSE	Relative squared error	no	0	∞	min	3
RRSE	Root relative squared error	no	0	∞	min	4
MAE	Mean absolute error	d	0	∞	min	5
RAE	Relative absolute error	no	0	∞	min	6
MAPE	Mean absolute percent. error	%	0	∞	min	7
NDEI	Non-dimensional error index	no	0	∞	min	8
r	Linear correlation coefficient	no	-1	1	close to 1	9
R^2	Coefficient of determination	%	0	∞	close to 100%	10
var(AE)	Variance of absolute errors	d^2	0	∞	min	11
var(APE)	Variance of absolute percentage errors	no	0	∞	min	12

$$MSE = \frac{1}{N}\sum_{i=1}^{N}(y_i - \hat{y}_i)^2 \tag{1}$$

$$RMSE = \sqrt{\frac{1}{N}\sum_{i=1}^{N}(y_i - \hat{y}_i)^2} \tag{2}$$

$$RSE = \frac{\sum_{i=1}^{N}(y_i - \hat{y}_i)^2}{\sum_{i=1}^{N}(y_i - avg(y))^2} \tag{3}$$

$$RRSE = \sqrt{\frac{\sum_{i=1}^{N}(y_i - \hat{y}_i)^2}{\sum_{i=1}^{N}(y_i - avg(y))^2}} \tag{4}$$

$$MAE = \frac{1}{N}\sum_{i=1}^{N}|y_i - \hat{y}_i| \tag{5}$$

$$RAE = \frac{\sum_{i=1}^{N}|y_i - \hat{y}_i|}{\sum_{i=1}^{N}|y_i - avg(y)|} \tag{6}$$

$$MAPE = \frac{1}{N}\sum_{i=1}^{N} \frac{|y_i - \hat{y}_i|}{y_i} * 100\% \tag{7}$$

$$NDEI = \frac{RMSE}{std(y)} \tag{8}$$

$$r = \frac{\sum_{i=1}^{N}(y_i - avg(y))\,(\hat{y}_i - avg(\hat{y}))}{\sqrt{\sum_{i=1}^{N}(y_i - avg(y))^2}\sqrt{\sum_{i=1}^{N}(\hat{y}_i - avg(\hat{y}))^2}} \tag{9}$$

$$R^2 = \frac{\sum_{i=1}^{N}(\hat{y}_i - avg(y))^2}{\sum_{i=1}^{N}(y_i - avg(y))^2} * 100\% \tag{10}$$

$$var(AE) = var(|y - \hat{y}|) \tag{11}$$

$$var(APE) = var(\frac{|y - \hat{y}|}{y}) \tag{12}$$

5 Results of Experiments

The performance of the models built by all nine algorithms for respective measures and for 15%, 30%, and 50% splits was presented in Fig. 3-14. For clarity, all measures were calculated for normalized values of output variables except for MAPE, where in order to avoid the division by zero, actual and predicted prices had to be denormalized. Almost in each instance the 50% split led to worse performance than the 15% split, what may be explained that the latter provided more data to learn models. In turn the 30% split gave various results, difficult to explain: they were once better and another time worse.

Fig. 11 shows there is high correlation, ranging from 61.1% to 83.7%, between actual and predicted prices for each model. In turn, Fig.12 illustrating the coefficients of determination indicates that from 50.7% to 82.6% of total variation in the dependent variable (prices) is accounted for by the models. This can be explained that data derived from the cadastral system and the register of property values and prices could cover only some part of potential price drivers.

The nonparametric Wilcoxon signed-rank tests were carried out for three measures: MSE, MAE, and MAPE and for each repeated split separately. The results are shown in Tables 3, 4, and 5 in the form of symmetric matrices, where a triple, eg. <035>, in each cell reflects the outcome for a given pair of models for 15%, 30% and 50% splits respectively. The numbers denote how many times there were statistically significant differences in mean values of respective errors within five repeated splits. Thus the triple <555> would denote that there were evidences against the H0 hypothesis in each instance, with the 0.05 significance level. However that result and <545> and <455> occurred in Tables 3, 4, and 5 only few times, majority of them for FOR models which outperformed CHI, ICA, and IEC models.

Due to the non-decisive results of majority of Wilcoxon tests, rank positions of individual algorithms were determined for each measure and each split (see Table 6). Observing median, average, minimal and maximal ranks it can be noticed that highest rank positions gained ANN, FOR, SVN, CRT, and KNN algorithms and the lowest IEC, ICA and CHI ones.

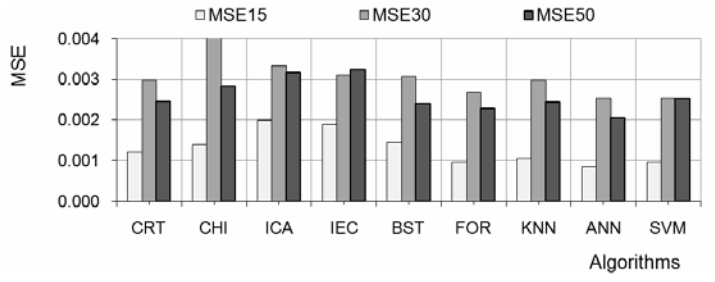

Fig. 3. Comparison of MSE values

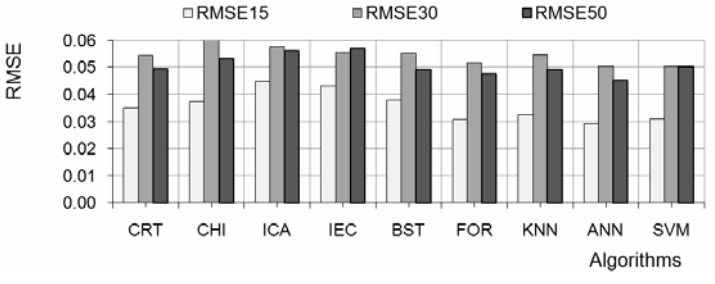

Fig. 4. Comparison of RMSE values

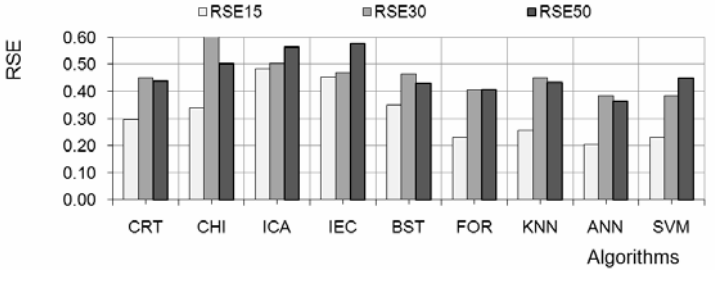

Fig. 5. Comparison of RSE values

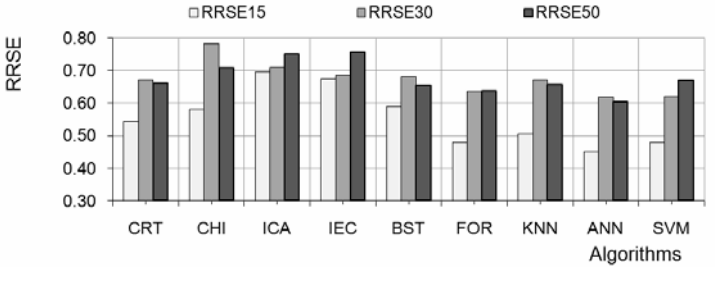

Fig. 6. Comparison of RRSE values

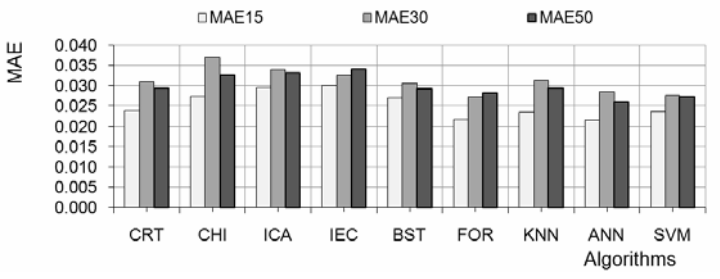

Fig. 7. Comparison of MAE values

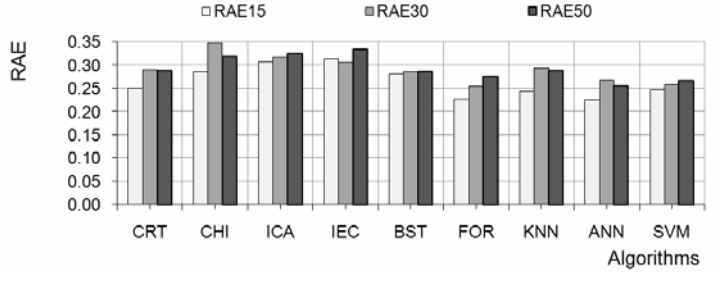

Fig. 8. Comparison of RAE values

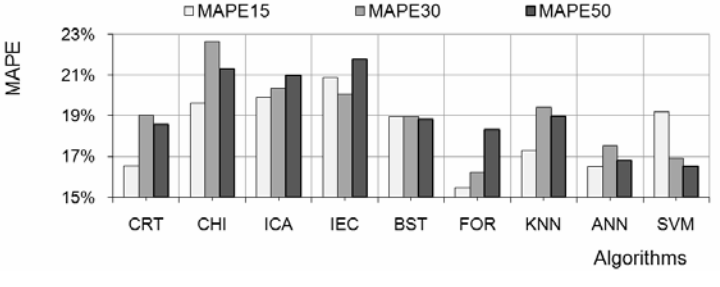

Fig. 9. Comparison of MAPE values

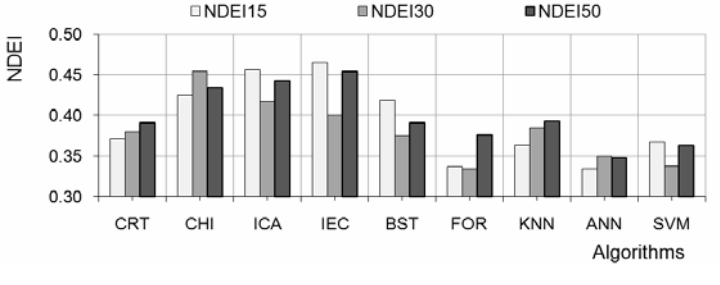

Fig. 10. Comparison of NDEI values

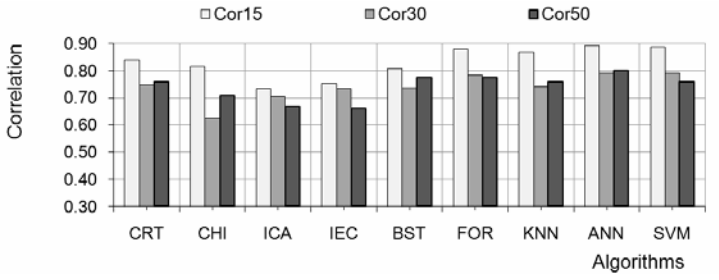

Fig. 11. Correlation between predicted and actual prices

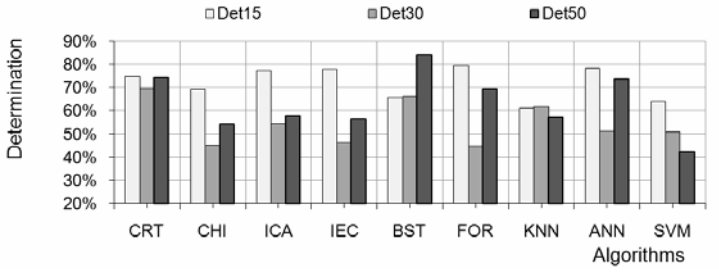

Fig. 12. Comparison of determination coefficient values

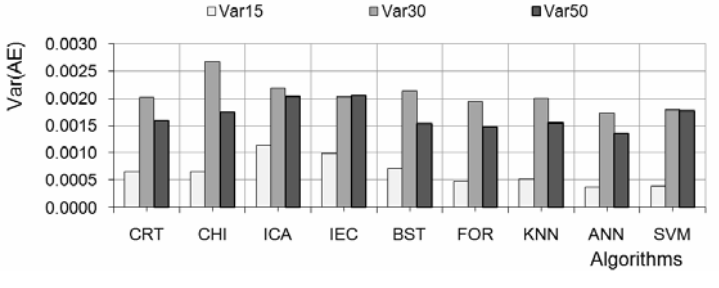

Fig. 13. Variance of absolute errors - Var(AE)

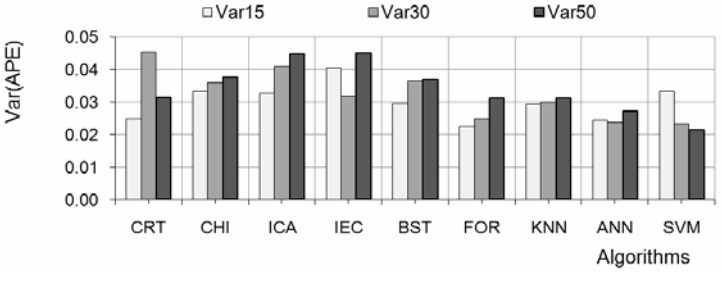

Fig. 14. Variance of absolute percentage errors - Var(APE)

Table 3. Results of Wilcoxon signed-rank test for squared errors comprised by MSE

	CRT	CHI	ICA	IEC	BST	FOR	KNN	ANN	SVM
CRT	-	145	232	332	300	224	103	023	012
CHI	145	-	151	144	345	545	243	555	344
ICA	232	151	-	011	334	555	110	344	133
IEC	332	144	011	-	243	555	121	344	123
BST	300	345	334	243	-	222	214	132	110
FOR	224	545	555	555	222	-	325	102	321
KNN	103	243	110	121	214	325	-	125	124
ANN	023	555	344	344	132	102	125	-	133
SVM	012	344	133	123	110	321	124	133	-

Table 4. Results of Wilcoxon test for absolute percentage errors comprised by MAPE

	CRT	CHI	ICA	IEC	BST	FOR	KNN	ANN	SVM
CRT	-	345	133	332	310	223	003	022	202
CHI	345	-	143	234	355	545	243	555	244
ICA	133	143	-	032	334	555	120	244	033
IEC	332	234	032	-	243	555	131	354	133
BST	310	355	334	243	-	221	214	202	132
FOR	223	545	555	555	221	-	335	211	303
KNN	003	243	120	131	214	335	-	115	334
ANN	022	555	244	354	202	211	115	-	332
SVM	202	244	033	133	132	303	334	332	-

Table 5. Results of Wilcoxon signed-rank test for absolute errors comprised by MAE

	CRT	CHI	ICA	IEC	BST	FOR	KNN	ANN	SVM
CRT	-	345	122	332	320	223	103	123	002
CHI	345	-	151	234	455	545	243	555	344
ICA	122	151	-	022	334	555	120	344	033
IEC	332	234	022	-	243	555	131	354	143
BST	320	455	334	243	-	222	214	102	130
FOR	223	545	555	555	222	-	235	202	301
KNN	103	243	120	131	214	235	-	115	324
ANN	123	555	344	354	102	202	115	-	233
SVM	002	344	033	143	130	301	324	233	-

Table 6. Rank positions of algorithms with respect to performance measures (1 means the best)

	Split	CRT	CHI	ICA	IEC	BST	FOR	KNN	ANN	SVM
MSE	15%	5	7	8	9	4	1	6	2	3
MSE	30%	4	5	9	7	8	1	3	6	2
MSE	50%	5	7	9	8	3	2	6	4	1
RMSE	15%	5	7	8	9	4	1	6	2	3
RMSE	30%	4	5	9	7	8	1	3	6	2
RMSE	50%	5	7	9	8	3	2	6	4	1
RSE	15%	4	7	9	8	5	1	6	3	2
RSE	30%	5	3	9	7	8	2	6	4	1
RSE	50%	7	5	8	9	4	3	6	2	1
RRSE	15%	4	7	9	8	5	1	6	3	2
RRSE	30%	5	3	9	7	8	2	6	4	1
RRSE	50%	7	5	8	9	4	3	6	2	1

MAE	15%	5	9	6	8	4	1	7	2	3
MAE	30%	6	5	9	7	8	1	4	2	3
MAE	50%	5	7	8	9	4	2	6	3	1
RAE	15%	4	8	7	9	5	1	6	2	3
RAE	30%	6	5	9	7	8	3	4	2	1
RAE	50%	4	8	7	9	5	2	6	3	1
MAPE	15%	5	9	7	8	3	1	4	2	6
MAPE	30%	6	5	9	7	8	2	4	1	3
MAPE	50%	4	8	7	9	5	3	6	2	1
NDEI	15%	5	7	8	9	4	1	6	2	3
NDEI	30%	4	5	9	7	8	1	3	6	2
NDEI	50%	5	7	9	8	3	2	6	4	1
r	15%	4	8	9	7	5	2	6	1	3
r	30%	3	6	9	7	8	1	5	4	2
r	50%	7	6	8	9	4	3	5	2	1
R^2	15%	3	9	7	5	2	6	4	1	8
R^2	30%	1	5	9	6	7	2	3	8	4
R^2	50%	3	9	8	9	4	1	7	6	2
Var(AE)	15%	6	7	8	9	4	3	5	2	1
Var(AE)	30%	3	5	9	7	8	1	4	6	2
Var(AE)	50%	5	8	9	6	3	2	4	7	1
Var(APE)	15%	4	6	8	9	3	5	7	1	2
Var(APE)	30%	6	2	9	7	9	3	5	4	1
Var(APE)	50%	5	8	7	9	6	3	4	1	2
median		5.00	7.00	9.00	8.00	5.00	2.00	6.00	2.50	2.00
average		4.69	6.39	8.31	7.86	5.33	2.00	5.19	3.22	2.14
min		1	2	6	5	2	1	3	1	1
max		7	9	9	9	9	6	7	8	8

6 Conclusions and Future Work

The experiments aimed to compare machine learning algorithms to create models for the valuation of residential premises which were conducted using the Statistica Data Miner. Nine methods comprising four algorithms to construct single regression trees: CRT, CHI, ICA, and IEC, two to build ensemble tree models: FOR and BST, and three others KNN, ANN, and SVM were applied to actual data sets derived from the cadastral system and the registry of real estate transactions. The overall conclusion is that ANN, FOR, SVN, CRT, KNN , and BST algorithms performed comparably well and the IEC, ICA and CHI algorithms should be rather discarded. Processing times of generating appraisal models using six best above mentioned algorithms, were not prohibitively long, therefore they can be incorporated into proposed internet system for real estate appraisals.

MAPE obtained in all tests ranged from 16.2% do 24.1%. This can be explained that data derived from the cadastral system and the register of property values and prices can cover only some part of potential price drivers. Physical condition of the premises and their building, their equipment and facilities, the neighbourhood of the building, the location in a given part of a city should also be taken into account, moreover, overall subjective assessment after inspection in site should be done. Therefore, we intend to test data obtained from public registers and then supplemented by experts conducting on-site inspections and evaluating more aspects of properties being appraised. What is more, further investigations of multiple models

comprising ensembles of different regression trees using bagging and boosting techniques is planned.

References

1. Breiman, L., Friedman, J., Olshen, R., Stone, C.: Classification and Regression Trees, Wadsworth Int. Group, Belmont (1984)
2. Torgo, L.: Functional Models for Regression Tree Leaves. In: Proc. 14th International Conference on Machine Learning, pp. 385–393. Morgan Kaufmann, San Francisco (1997)
3. Badulescu, L.A., Nicula, A.: Data Mining Decision Trees in Economy. Analele Universitatii din Oradea, Stiinte Economice XVI II, 723–727 (2007)
4. Giudici, P., Figini, S.: Applied Data Mining for Business and Industry. Wiley, Chichester (2009)
5. Trana, V.T., Yanga, B.S., Oha, M.S., Tanb, A.C.C.: Machine condition prognosis based on regression trees and one-step-ahead prediction, Mech. Systems and Signal Process. 22(5), 1179–1193 (2008)
6. Khoshgoftaar, T.M., Seliya, N.: Tree-based software quality estimation models for fault prediction. In: Proc. 8th IEEE Symp. on Software Metrics, pp. 203–214 (2002)
7. Wang, X., Dietterich, T.G.: Efficient Value Function Approximation Using Regression Trees. In: Proceedings of the IJCAI Workshop on Statistical Machine Learning for Large-Scale Optimization (1999)
8. Podgorelec, V., Kokol, P., Stiglic, B., Rozman, I.: Decision trees: an overview and their use in medicine. Journal of Medical Systems 26, 445–463 (2002)
9. Balac, N., Gaines, D.M., Fisher, D.: Using Regression Trees to Learn Action Models. IEEE Int. Conference on Man and Cybernetics 5, 3378–3383 (2000)
10. Acciani, C., Fucilli, V., Sardaro, R.: Model Tree: An Application in Real Estate Appraisal, 109th EAAE Seminar, no. 44853, Viterbo (2008)
11. Gokhale, S.S., Lyu, M.R.: Regression tree modeling for the prediction of software quality. In: Proc. of ISSAT 1997, pp. 31–36 (1997)
12. Perlich, C., Provost, F., Simonoff, J.S.: Tree induction vs. logistic regression: a learning-curve analysis. J. of Machine Learning Res. 4, 211–255 (2003)
13. Quinlan, J.R.: Learning with Continuous Classes. In: 5th Australian Joint Conference on Artificial Intelligence (AI 1992), Singapore, pp. 343–348 (1992)
14. Król, D., Lasota, T., Trawiński, B., Trawiński, K.: Investigation of evolutionary optimization methods of TSK fuzzy model for real estate appraisal. International Journal of Hybrid Intelligent Systems 5(3), 111–128 (2008)
15. Lasota, T., Pronobis, E., Trawiński, B., Trawiński, K.: Exploration of Soft Computing Models for the Valuation of Residential Premises using the KEEL Tool. In: Nguyen, N.T., et al. (eds.) 1st Asian Conference on Intelligent Information and Database Systems (ACIIDS 2009), pp. 253–258. IEEE, Los Alamitos (2009)
16. Hill, T., Lewicki, P.: Statistics: Methods and Applications. StatSoft, Tulsa (2007)
17. Breiman, L.: Random Forests. Machine Learning 45, 5–32 (2001)
18. Hagquist, C., Stenbeck, M.: Goodness of Fit in Regression Analysis – R^2 and G^2 Reconsidered. Quality & Quantity 32, 229–245 (1998)
19. Witten, I.H., Frank, E.: Data Mining: Practical machine learning tools and techniques, 2nd edn. Morgan Kaufmann, San Francisco (2005)